

Respiratory Emergencies (10/5)

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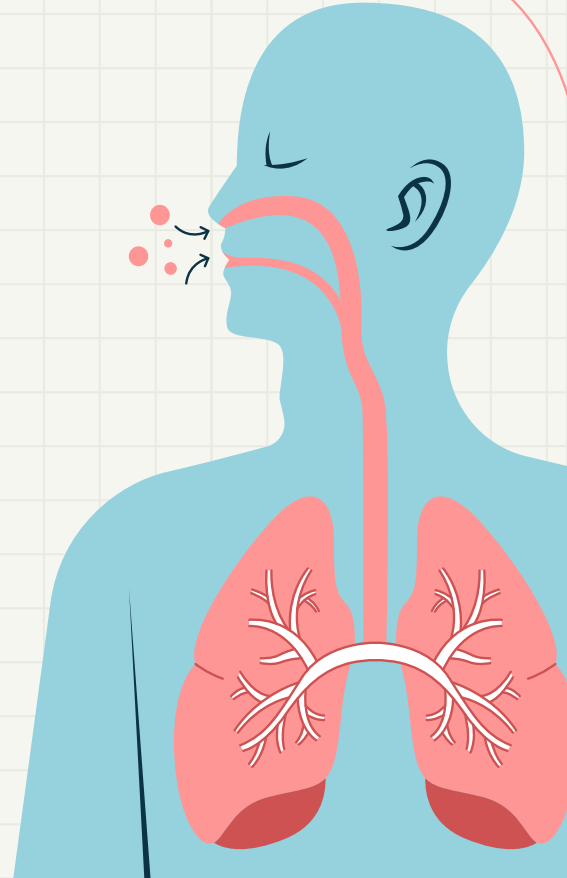




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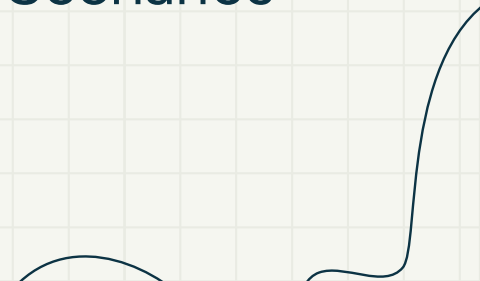
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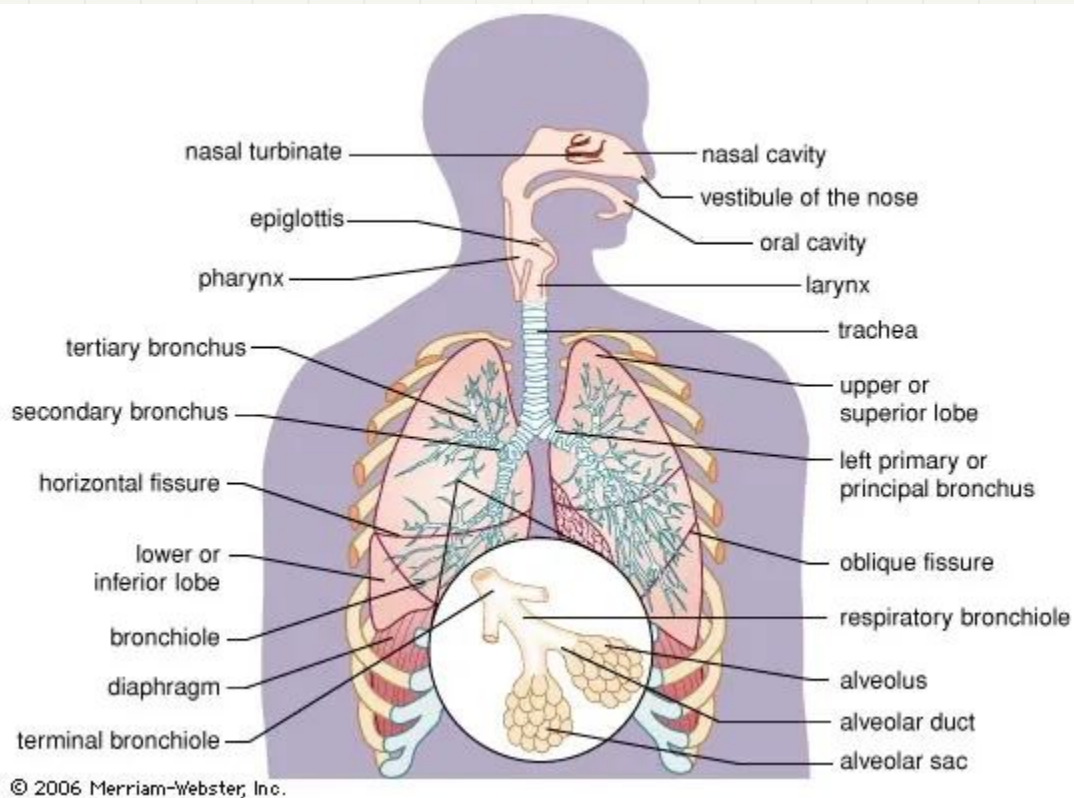




01

Anatomy of the Respiratory system

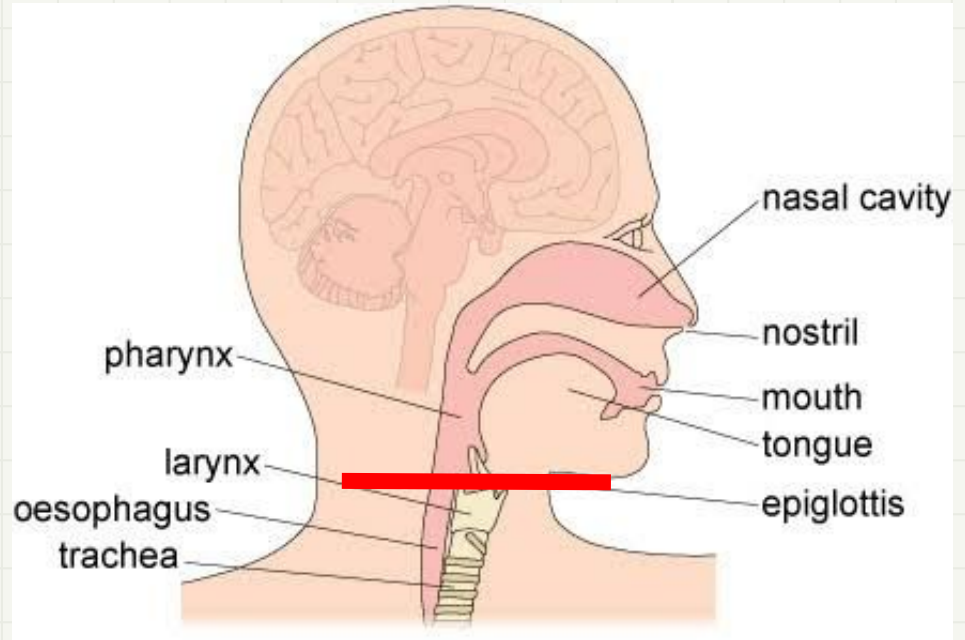
Overview



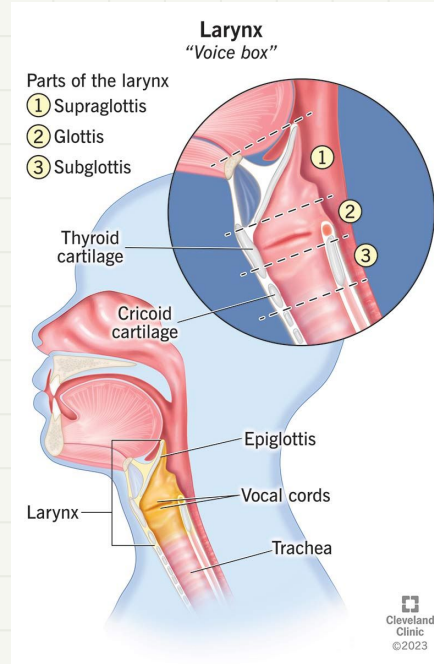
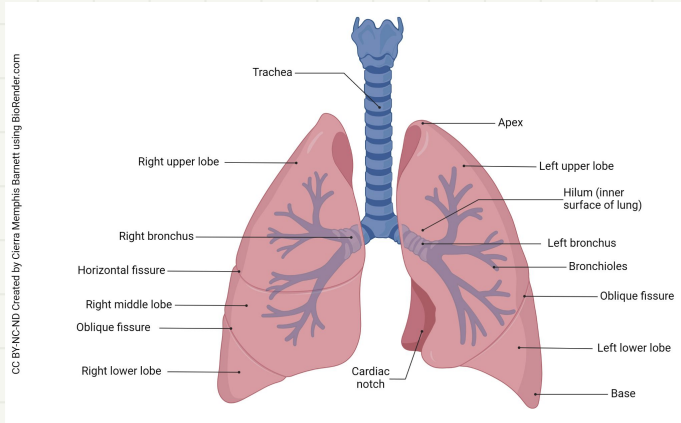
Upper Airway

Main Sections of the Upper Airway

- Nasal Cavity
- Oral Cavity
- Pharynx
- Larynx (Kind of)



Lower Airway



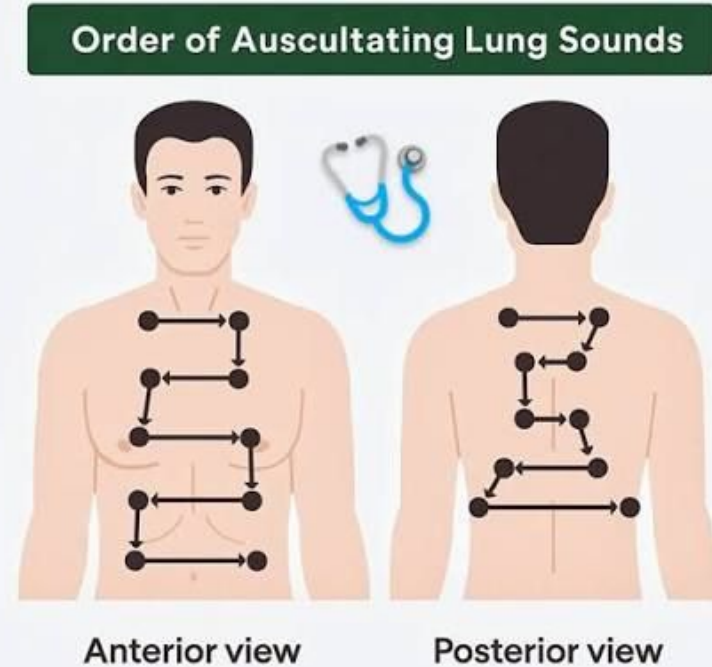
Main Sections of the Lower Airway

- Larynx
- Trachea
- Carina
- Bronchi
- Bronchioles
- Alveoli

Lung Sounds

Adventitious (Abnormal) Lung Sounds

- **Wheezing** → Lower airway constriction
- **Rales** → Fluid in the lungs
- **Rhonchi** → Mucus/secretion
- **Stridor** → Obstruction of upper airway



Important interventions

NASOPHARYNGEAL AIRWAY (Nasal Airway "Nasal Trumpet")

A soft, flexible tube inserted through the nose into the nasopharynx to maintain a patent airway.



INDICATIONS

- Semiconscious patient with intact gag reflex
- Oral or jaw trauma making oral airway insertion difficult
- Facilitates nasal suctioning or positive pressure ventilation

INSERTION TECHNIQUE

- Size from nose tip to earlobe; lubricate with water-soluble
- Insert gently with bevel toward septum
- Advance along nasal floor until flange rests at nostril
- Confirm patency by assessing ventilation or airflow
- Do not force if obstruction met; try other nostril

CONTRAINDICATIONS

- Severe nasal or midface trauma
- Signs of skull base fracture, CSF rhinorrhea
- Nasal obstruction, nasal infection

@dr.curative

The Oropharyngeal Airway (OPA): A Quick Guide

When and How to Use an OPA



Prevents Airway Obstruction

It holds the tongue down to keep the airway clear for breathing.



For Unconscious Patients ONLY

Intended for patients without a gag reflex to secure an open airway.



Correct Sizing is Crucial

Measure from the corner of the patient's mouth to the angle of the jaw.

Critical Contraindications (When NOT to Use)



DO NOT use on conscious or semi-conscious patients
It can trigger the gag reflex, causing vomiting and aspiration.

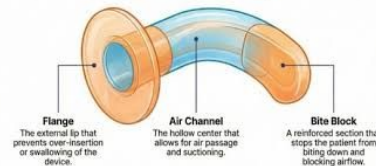


Use a Nasopharyngeal Airway (NPA) for conscious patients instead



Avoid with significant oral trauma or bleeding
Can worsen injuries and cause complications.

Anatomy of the OPA



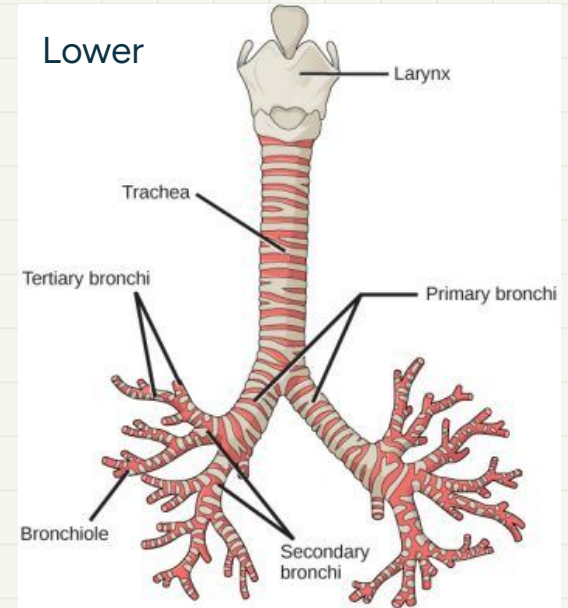
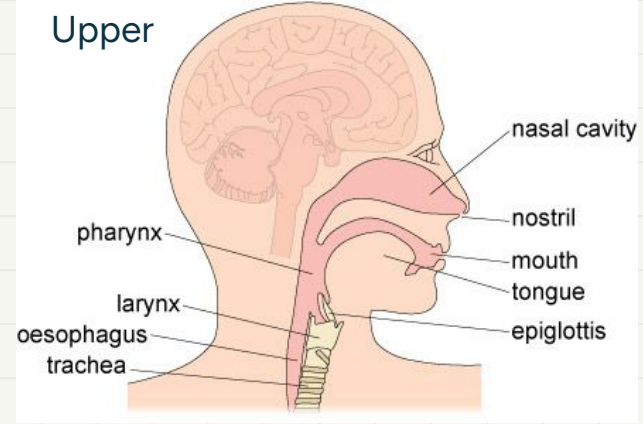


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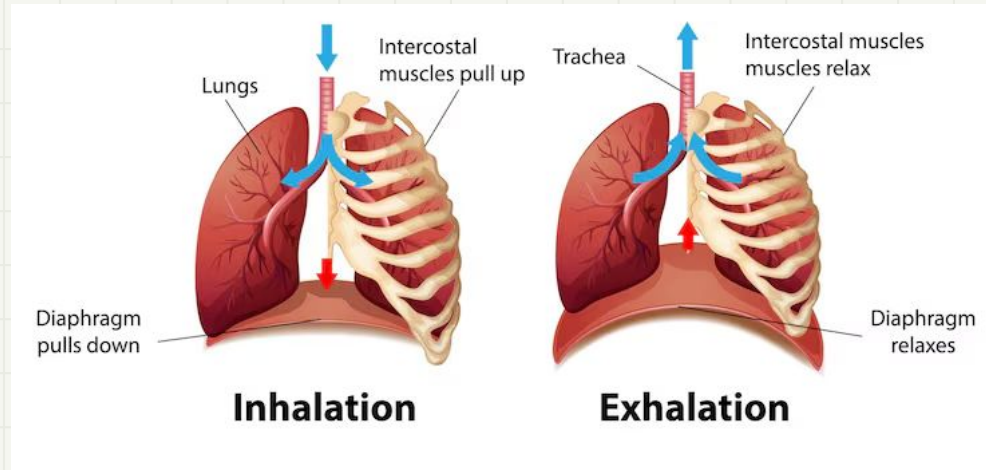
Physiology of the Respiratory system

Respiratory Tract

- Air enters through the nose or mouth.
- It travels through the pharynx and larynx.
- Air moves down the trachea into the bronchi and bronchioles.
- The air eventually reaches the alveoli for gas exchange



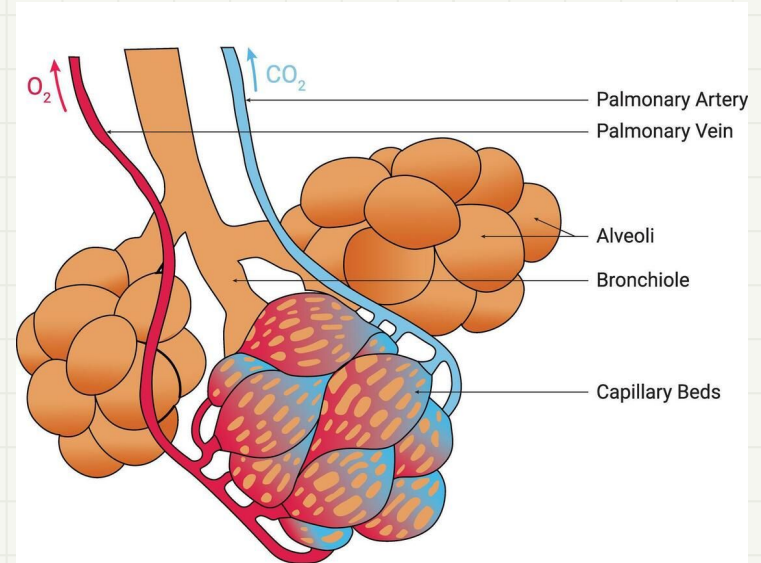
How Breathing Works:



- The diaphragm contracts and moves down during inhalation.
- This increases chest space, decreasing pressure, allowing air to enter the lungs.
- The diaphragm relaxes and moves up during exhalation.
- This decreases chest space, increasing pressure, pushing air out of the lungs.

Alveoli & Gas Exchange

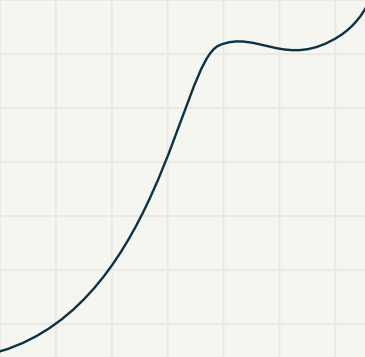
- Alveoli have thin walls and are surrounded by capillaries
- Oxygen moves from the alveoli into the blood.
- Carbon dioxide moves from the blood into the alveoli.
- Gas exchange happens through the thin alveolar walls.
- CO₂ is exhaled, while oxygen is carried throughout the body.





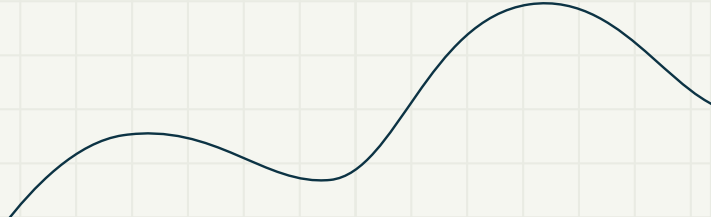
03

**What is your Field
Impression?**



Scenario 1

35 year old male who is in tripod position and visibly anxious.



Correct Answer:
Asthma Attack

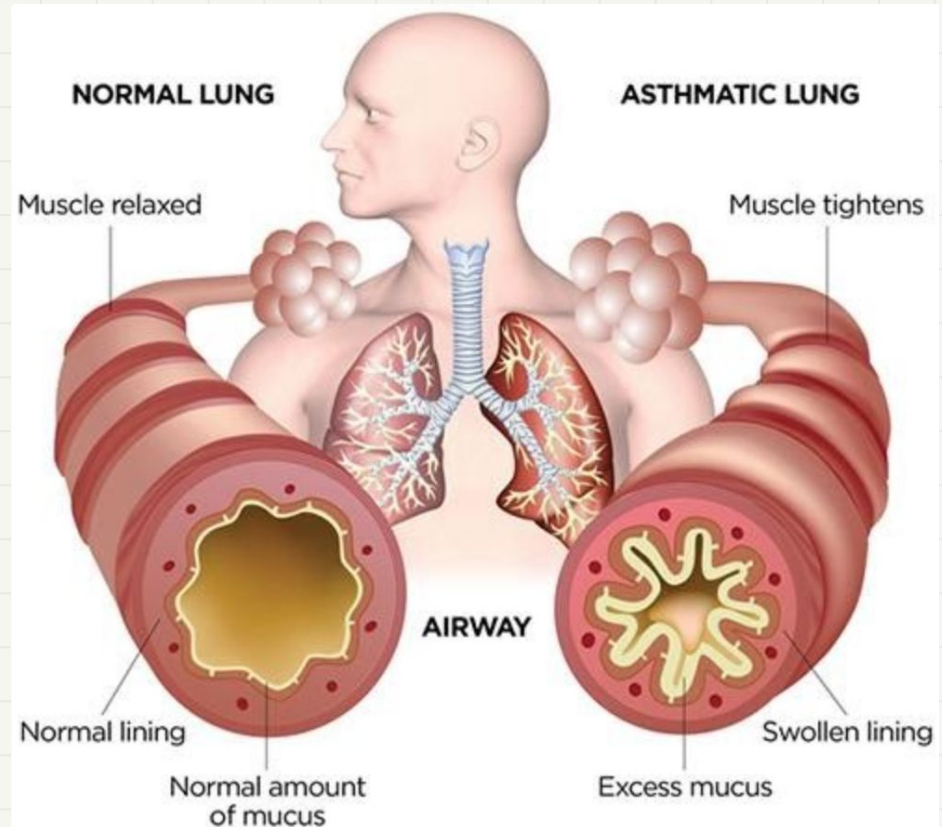
Asthma Attack

What is Epiglottitis

- Narrowing of the airways
- Airway inflammation and swelling
- Excess mucus can block airflow

Clinical Presentation

- Shortness of breath
- Wheezing
- Chest tightness
- Rapid, labored breathing



EMT Actions

- ❖ Keep patient sitting upright
- ❖ Administer oxygen as needed
- ❖ Assist with Albuterol
- ❖ Monitor SpO₂ and lung sounds
- ❖ Reassess frequently

ASTHMA ATTACK

signs & symptoms and management

The infographic illustrates the signs and symptoms of an asthma attack and provides management steps. It features a central illustration of a woman using an inhaler, with arrows pointing to various symptoms. A circular inset compares a normal airway with an asthma attack airway, showing narrowed airways and swollen muscles.

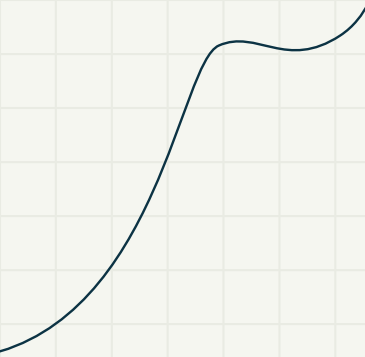
Signs & Symptoms:

- Wheezing:** A whistling sound may happen while breathing out.
- Shortness of breath:** Breathing may feel difficult or not enough.
- Chest tightness:** The chest may feel tight, heavy, or squeezed.
- Coughing:** Cough may worsen at night, with exercise, or during triggers.
- Severe tiredness / drowsiness:** Extreme sleepiness, confusion, or exhaustion is a danger sign.
- Blue lips warning:** Blue lips, grey face, or severe breathlessness needs urgent help.
- Trouble speaking:** Speaking full sentences may become difficult.
- Fast breathing:** Breathing may become rapid or shallow.

MANAGEMENT

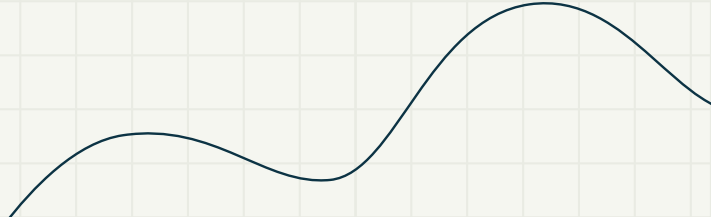
- Sit upright:** Sit up straight and try to stay calm. Do not lie down.
- Use reliever inhaler:** Use the prescribed quick-relief inhaler as directed.
- Use a spacer if available:** A spacer helps more medicine reach the lungs.
- Follow asthma action plan:** Use your written plan for worsening symptoms.
- Avoid triggers:** Move away from smoke, dust, cold air, pets, or strong smells if possible.
- Seek emergency help:** Get urgent care if breathing is severe, lips turn blue, or inhaler is not helping.

An asthma attack can become serious quickly. Use your prescribed reliever inhaler and seek urgent help if symptoms are severe or not improving.



Scenario 2

5 year old child with high fever
and difficulty swallowing



Correct Answer: Epiglottitis

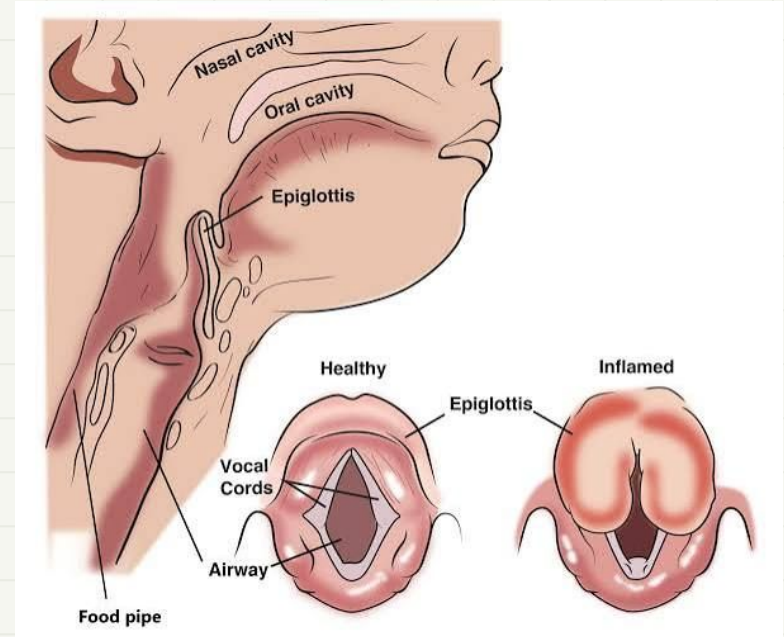
Epiglottitis

What is Epiglottitis

- Swelling of the epiglottis
- Is present in children
- Bacterial Infection

Clinical Presentation

- Tripod Position
- Inspiratory stridor
- Rapid onset high fever



EMT Actions

Supplement O2 Via Blow By

Position of Comfort

Have patient remain calm

Request ALS

ACUTE EPIGLOTTITIS:
Life-threatening inflammation & airway obstruction
By X @SaleemHamilah



airway obstruction.

♦ **ETIOLOGY**

- ✓ **Most common cause:** *Haemophilus influenzae type B (Hib)* (↓ incidence due to vaccination)
- ✓ **Others:** *Streptococcus spp.*, *Staph aureus*

♦ **AGE GROUP**

Typically children 2–7 years, but also seen in **unvaccinated adults**

♦ **CLINICAL FEATURES (Mnemonic: “4 D’s + S”)**

- ✓ **DYSPHAGIA**
- ✓ **DROOLING**
- ✓ **DYSPHONIA** (muffled “hot potato” voice)
- ✓ **DYPNEA**
- ✓ **STRIDOR** (inspiratory), tripod positioning, high fever, toxic appearance

♦ **DIAGNOSIS**

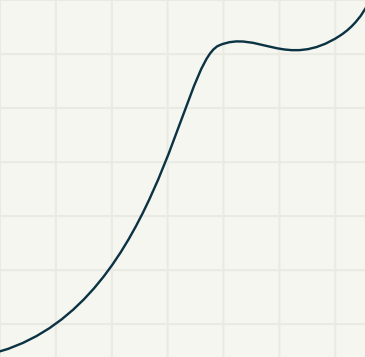
- ✓ **Clinical diagnosis**—do **NOT** delay treatment ⚠
- ✓ **Lateral neck X-ray:** “Thumbprint sign” (enlarged epiglottis)
- ✓ **Definitive: Direct laryngoscopy** in OR setting (only if airway is secure)



♦ **MANAGEMENT**

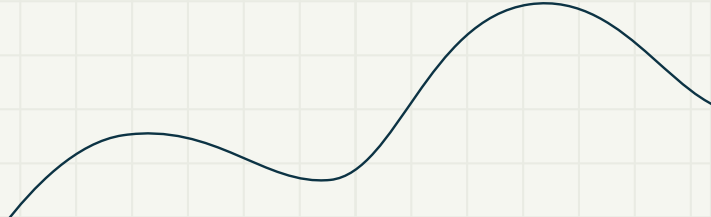
✓ **AIRWAY PROTECTION FIRST!**

- Intubation in OR if respiratory distress
- ✓ **Empiric IV antibiotics:** Ceftriaxone or cefotaxime + vancomycin
- ✓ **IV corticosteroids** (e.g., dexamethasone) may reduce inflammation
- ✓ **Hib vaccination** = prevention



Scenario 3

22 year old male with sudden chest pain and difficulty breathing.



Correct Answer:
Spontaneous Pneumothorax

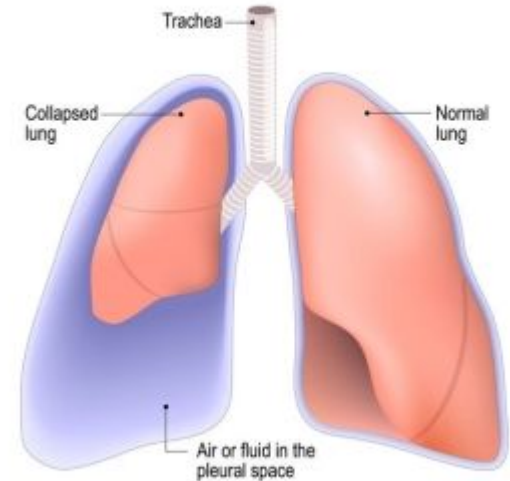
Spontaneous Pneumothorax

What is it?

- The lung collapses spontaneously
- Air leaks out the lungs into the pleural cavity
- Trapped air puts pressure on the lungs, preventing full expansion

Clinical Presentation

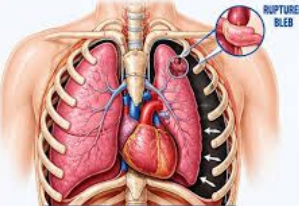
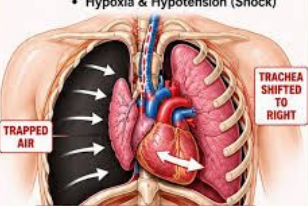
- Tripod Position
- Sudden chest pain
- Rapid breathing
- Decreased breath sounds on one side



EMT Actions

- ❖ Oxygen
- ❖ Position of comfort
- ❖ Monitor SpO₂ & lung sounds
- ❖ Frequent reassessment
- ❖ Rapid transport

PNEUMOTHORAX EMERGENCY: TYPES & TRIAGE
(NCLEX RESPIRATORY & CRITICAL CARE HIGH-YIELD HACK)

1. SPONTANEOUS PNEUMOTHORAX	2. TENSION PNEUMOTHORAX
 Definition: Rupture of a pulmonary bleb/bleb (air-filled sac) on the lung surface.	 Definition: Air enters the pleural space on inspiration but CANNOT escape on expiration.
 Key Signs: Sudden, sharp chest pain & dyspnea (shortness of breath).	 CRITICAL EMERGENCY: Trapped air rapidly increases pressure, collapsing the lung AND compressing the heart/vessels.
 Assessment Findings: • ↓ Breath sounds on affected side • Hyperresonance on percussion • Normal Trachea position	 Assessment Findings: • TRACHEAL DEVIATION (Shift to unaffected side) • JVD (Jugular Vein Distention) • Hypoxia & Hypotension (Shock)
 RUPTURED BLEB LUNG PARTIALLY COLLAPSED HEART REMAINS CENTERED	 TRAPPED AIR TRACHEA SHIFTED TO RIGHT LUNG COLLAPSED COMPLETELY HEART & TRACHEA SHIFTED
 NCLEX EMERGENCY PRIORITY ACTION: For a Tension Pneumothorax, do NOT wait for a Chest X-ray! The immediate, life-saving intervention is an urgent NEEDLE DECOMPRESSION (14-gauge needle into 2nd intercostal space) to release trapped air, followed by Chest Tube insertion!	
SAVE THE PNEUMOTHORAX MATRIX! TAG YOUR STUDY SQUAD	

Kahoot!

